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Bond Strength of Cement Grouted Glass Fibre Reinforced Plastic (GFRP) Anchor Bolts

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Fibre reinforced plastic (FRP) has recently been introduced in the market in the form of bars for grouted anchor bolts. The resistance to corrosion and chemical attacks, high strength-to-weight ratio, low electromagnetic properties, and ease in handling of these bars make them a better alternative to steel in some applications of grouted anchor bolts. However, to fully utilise FRP bars as tendons for cement grouted anchors, some aspects of their behaviour have to be determined, including load carrying capacity, bond strength in cement grout, long-term strength and durability in alkaline environments. In this paper, the load carrying capacity and bond strength of cement grouted glass fibre reinforced plastic (GFRP) anchor bolts are discussed in comparison with steel anchor bolts. The results of laboratory and field pull-out tests of GFRP and steel bars anchored with cement grout are presented. The pull-out tests were conducted on four types of GFRP bars and two types of steel bars installed in concrete blocks and rock mass. The experimental results have shown that the bond strength of GFRP bar anchor bolts is close to that of steel bar anchor bolts. The slip at failure of GFRP bars relative to the cement grout is greater than that of steel bars, mainly due to their low modulus of elasticity.

INTRODUCTION

The use of grouted anchor bolts continues to become more common in civil and mining engineering. They are being used both as temporary and permanent structural members to ensure the stability of structures such as slopes, retaining walls, bridge abutments, tunnels, underground excavations, reinforced concrete foundations, lock walls, etc. Grouted anchors are also used for the strengthening and rehabilitation of existing structures [1, 2]. A grouted anchor bolt is, in a general term, a bar, which is inserted in a hole drilled in rock/concrete and then grouted [3]. Steel has been used as anchor tendons for many years.

In the last few years, fibre reinforced plastic (FRP) bars have been introduced in the market as tendons for grouted anchors and as reinforcement for concrete structures [4–11]. Aramid FRP (AFRP), Carbon FRP (CFRP) and Glass FRP (GFRP) bars are the currently available products. FRP bars present many advantages such as resistance to corrosion and chemical attack, high strength-to-weight ratio and electro-magnetic neutrality. These advantages could improve the durability of anchor bolts and facilitate transportation, handling and installation. Besides, optical fibre sensors can be integrated into FRP bars during the fabrication process for permanent monitoring of anchor bolts. However, only limited experimental information is available about FRP bars as tendons for grouted anchor bolts. Performance characteristics such as bond strength with grout, pullout performance, long-term strength and resistance to alkaline environments (cement grout) have to be determined. This paper investigates the bond strength behaviour and load carrying capacity of cement grouted GFRP anchor bolts.

GFRP bars are available in nominal diameters similar to those of conventional and post-tensioning steel bars, however, they differ from steel bars in two important aspects [12]. First, the pultrusion process by which the GFRP bars are manufactured results in bars with a fairly constant cross-section and a smooth surface. Indentations are produced on the bar by wrapping a fibre glass string around the bar before the resin sets. These indentations are different from the deformation in steel bars and generally shallower. Moreover, some manufacturers

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Table 1. Physical and mechanical properties of commercial glass fibres

Parameter	E-glass	S-glass	C-glass	AR-glass
Tensile strength (GPa)	3.45	4.3	3.03	2.5
Tensile modulus (GPa)	72.4	86.9	69.0	70.0
Ultimate strain (%)	4.8	5.0	4.8	3.6
Poisson's ratio	0.2	0.22	—	—
Density (g/cm^3)	2.54	2.49	2.49	2.78
Diameter (μm)	10.0	10.0	4.5	—
Longitudinal CTE (10^{-6}/°C)	5.0	2.9	7.2	—
Dielectric constant	6.3	5.1	7.2	—

add a coating of sand and resin on the bar at the end of the manufacturing process that makes the bar surface even smoother. Second, the stress–strain relation of GFRP bars is linear at all stress levels up to the point of failure, without exhibiting any yielding of the material. The modulus of elasticity of GFRP bars is approximately 20–25% of that of steel. The tensile strength varies from 500 to 1100 MPa, depending on the glass content, type of fibre and resin, and manufacturing process [6].

Since the surface deformation and the mechanical properties of FRP bars are different from those of steel bars, sufficient experimental information about bond properties and load carrying capacity of grouted FRP anchor bolts is required to develop design guidelines [13].

RESEARCH SIGNIFICANCE AND OBJECTIVES

The objective of the project reported in this paper was to study the bond properties and the pull-out performance of cement grouted GFRP anchor bolts. Load carrying capacity, bond strength, load–slip relation and critical bond length were obtained from the laboratory and field pull-out tests of grouted GFRP anchor bolts. Pull-out tests of cement grouted conventional steel bar and threaded Dywidag bar anchor bolts were also conducted for comparison purposes. The experimental programme included four different types of GFRP bars with a similar nominal diameter of 25 mm. One conventional cement grout was used for all the tests. Large concrete blocks were used for the laboratory tests, while a rock mass was selected for the field tests.

FRP COMPOSITE MATERIALS

A composite material can be defined as a macroscopic combination of two or more distinct materials with recognisable interfaces [14]. FRP composite materials are essentially composed of filaments in a resin matrix. The filaments consist of various kinds of load bearing fibres of high tensile strength and high modulus of elasticity. The most commonly used fibres are aramid, carbon and glass [6]. The matrix functions as a bonding material to hold the fibres together so as to allow load transfer and prevent shear between individual fibres, protect the fibres and maintain the dimension stability. The most commonly used matrices are polyester, epoxy and vinyl ester resins [6]. GFRP bars used in this study consist of glass fibres impregnated in thermosetting polyester resin. A brief description of GFRP constituent materials is given below.

Glass fibres

Glass fibres are the most common of all reinforcing fibres for FRPs and are fabricated by extruding molten glass through an orifice [7]. Their principal advantages are low cost, high tensile strength and excellent insulating properties. The two types of most commonly used glass fibres are E-glass and S-glass. E-glass fibre has the lowest cost of all commercially available reinforcing fibres, and is used for a general purpose where strength, electrical resistance, acid resistance and low cost are important. S-glass fibre has higher strength, stiffness and ultimate strain, but is more expensive and susceptible to degradation in alkaline environments than E-glass [15]. Other types of glass are C and alkali-resistant (AR). C-glass fibre is used for its chemical stability in acidic environments. AR-glass is developed to minimise weight and strength loss in alkaline environments [15]. The physical and mechanical properties of glass fibres are shown in Table 1.

Polyester resin

Thermoset polyester resins usually consist of an unsaturated ester polymer dissolved in a crosslinking monomer such as styrene [16]. Depending on the mix of ingredients, the properties of polyester resins may vary widely. Polyester resins are resistant to fire, moisture, acids and alkalies, but are degradable by chlorinated solvents [16, 17]. The upper use temperature of polyester is 120°C [16]. The main advantages of polyester for FRPs are low viscosity, fast cure time, dimension stability, excellent chemical resistance and moderate cost [18]. The disadvantage of unfilled polyesters is their high volumetric shrinkage during processing. However, the combination of low cost, excellent properties and processability makes it the most widely used resin for FRPs. The physical and mechanical properties of polyester resin are shown in Table 2.

Table 2. Physical and mechanical properties of polyester resin [18]

Parameter	Polyester resin
Tensile strength (MPa)	20–100
Tensile modulus (GPa)	2.1–4.1
Ultimate strain (%)	1–6
Density (g/cm^3)	1.0–0.45
Thermal expan. coefficient (10^{-6}/°C)	55–100
Cure shrinkage (%)	5–12

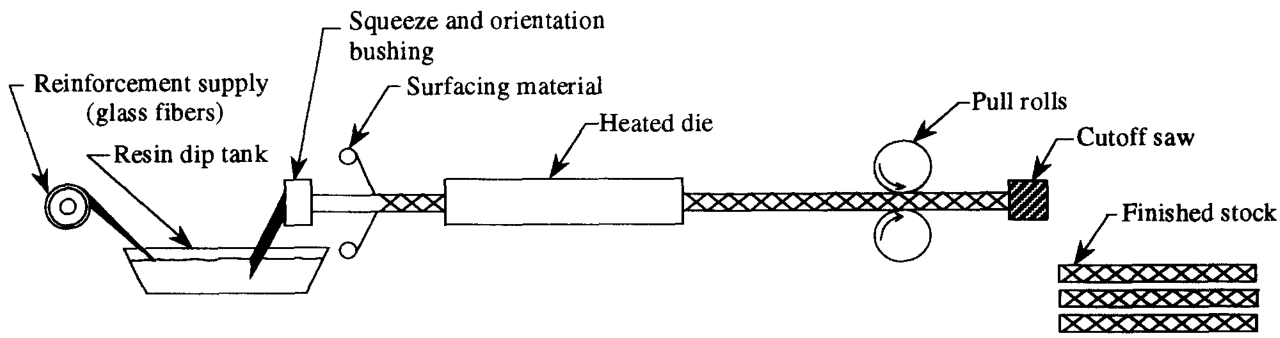

Fig. 1. Pultrusion process of GFRP bars.

DESCRIPTION AND PROPERTIES OF GFRP BARS USED IN THE STUDY

GFRP bars are usually manufactured using a pultrusion process as shown in Fig. 1. This kind of process allows high-fibre content products to be obtained, 60–80% by volume, with a homogeneous distribution of fibres in the cross-section of the bar. The ESM images presented in Fig. 2 show the high fibre content of one of the GFRP bars used in this study (approximately 73% by volume). The images also show that fibres used for the bar have a diameter of about 15 μm and the homogeneity of fibre distribution across the section of the bar.

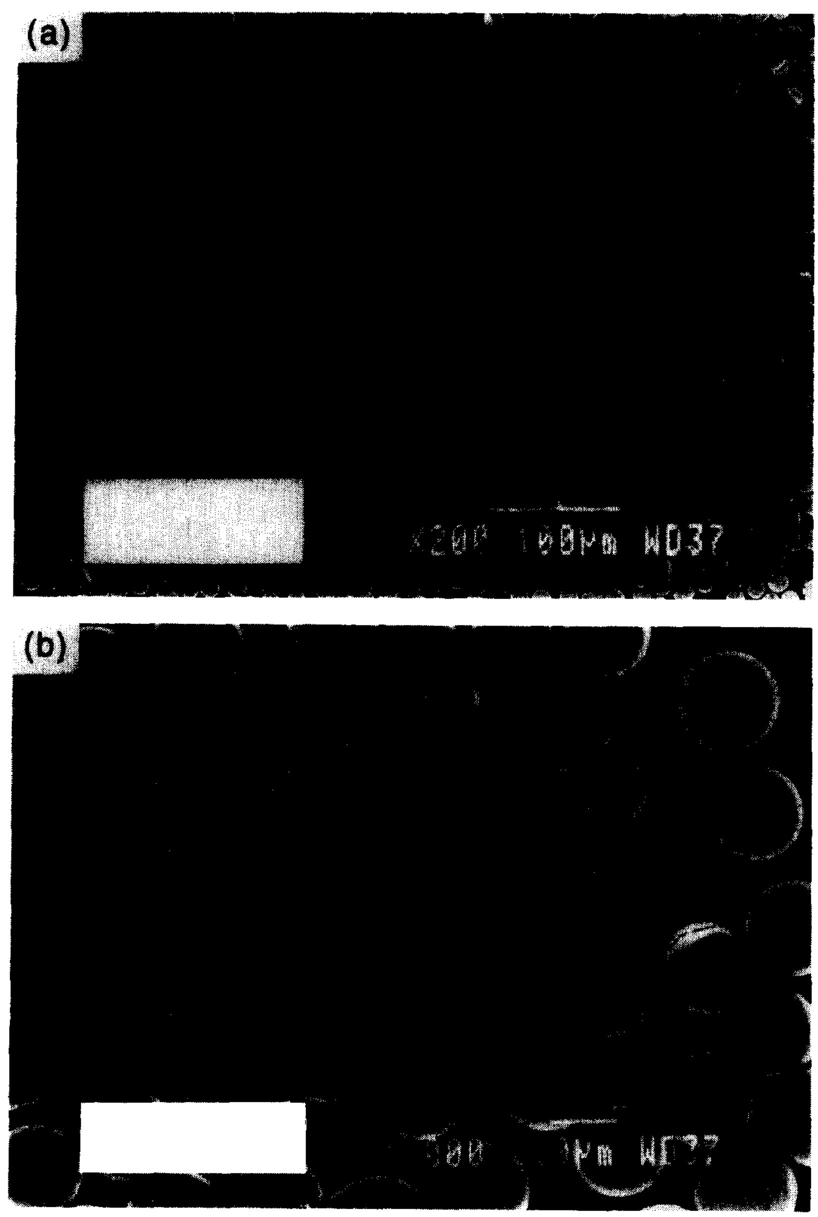

Fig. 2. Cross-section image of resin bonded glass fibre bar (electron scan microscope).

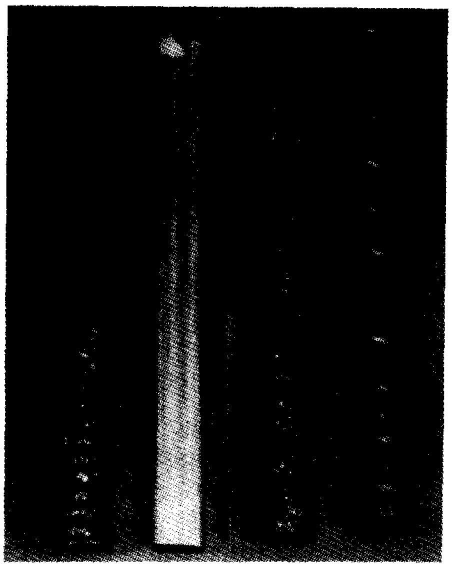

Fig. 3. Overview of GFRP bars in the study. (from left to right: A, C, B-2 and B-1).

Four types of GFRP bars, namely GFRP A, B-1, B-2 and C with a nominal diameter of 25 mm, were selected for the tests. Conventional steel threaded bars and threaded Dywidag post-tensioning bars of equal diameter were also used for comparison purposes.

An overview of the GFRP bars used is shown in Fig. 3. All four types of GFRP bars are composed of continuous longitudinal E-glass fibres impregnated in a thermoset polyester resin and manufactured using a pultrusion process. GFRP A is manufactured by a German company (Dywidag Systems International, DSI) as rock bolts. The surface tows of GFRP A are stressed using the same type of fibres as the bar during the manufacturing process so that indentations are obtained. GFRP B-1 and B-2 are manufactured by an American company (International Grating Inc., IGi) as concrete reinforcement as well as rock bolts. The GFRP B-1 bar includes a small quantity of fibres which are wrapped around the longitudinal fibres in a spiral pattern. The function of the wrapping is to induce deformation on the surface of the bar. The depth and pitch of the indentation of GFRP B-1 are much larger than those of GFRP A. GFRP B-2 is obtained from GFRP B-1 by adding a layer of sand and resin coating on the bar at the end of the manufacturing process. GFRP C is manufactured by a French company (Doneco Ferrettite Celtite, DFC) as anchor bolts. This bar has a smooth surface and fibres on the surface of the bar are loose.

Table 3 shows the mechanical properties of GFRP and steel bars. It can be seen that the tensile strength and modulus of elasticity of GFRP A, B-1/B-2 and C are 958 MPa and 50 GPa, 690 MPa and 45 GPa, and 595 MPa and 39 GPa, respectively. The tensile strength of GFRP bars is 1.5–2.4 times that of conventional steel bars and 0.7–1.1 times that of Dywidag bars. The modulus of elasticity of GFRP bars is 20–25% that of steel bars. For the tensile strength of the bars shown in Table 3, the corresponding ultimate loads are 470, 339 and 292 kN for GFRP A, B-1/B-2 and C, respectively. The conventional steel bar and Dywidag bar of 25 mm have elastic limit and ultimate loads of 196 and 295 kN, and 417 and 515 kN, respectively.

EXPERIMENTAL PROGRAMME

Laboratory pull-out tests

Large concrete blocks of 600 × 600 × 400 mm were cast as a medium to install anchor bolts. The compressive strength and modulus of elasticity of the concrete were 83 MPa and 38 GPa, respectively. Each block hosted four bolts. Holes of 50.8 mm diameter were cast using polymer cylinders as formers in the blocks to achieve a more realistic degree of drillhole wall roughness, which can prevent the failure of anchor bolts at the concrete and grout interface. The polymer cylinders were pulled out two days after casting. Two bond lengths of 63.5 mm (2.5 d_b) and 127 mm (5 d_b) were selcted for the anchor bolts, except for GFRP C anchor bolts which only had a bond length of 127 mm (5 d_b), where d_b is the diameter of the bar. The top part, 150 mm above the bond length, was isolated from the grout using a PVC tube to eliminate the effect of support reaction on the bolt. The spacing between anchorage units in each concrete block was kept at a minimum of 180 mm to avoid any effects (stress superposition, crack propagation, etc.) due to adjacent anchor bolts. Four pull-out tests were conducted for each type of bolt 14 days after injection.

All anchor bolts were injected with a conventional cement grout made from Type 10 Portland cement (ASTM I) with a water–cement (W/C) ratio of 0.4. At the time of testing, the grout had developed a compressive strength of 52 MPa and modulus of elasticity of 17 GPa. Pull-out tests on the anchor bolts were carried out using a hollow hydraulic jack having a capacity of 300 kN. The pull-out load and slip at the loaded end were measured using a 500 kN load cell and a Linear Voltage Differential Transformer (LVDT), respectively. The load cell and the LVDT were connected to a data logger system. The rate of loading was fixed at 5 kN/sec.

Field pull-out tests

Field pull-out tests were conducted to determine the pull-out performance of GFRP anchor bolts installed

Table 3. Mechanical properties of GFRP and steel bars

Parameter	GFRP A	GFRP B-1/B-2	GFRP C	Steel	Dywidag
Tensile strength (MPa)	958	690	595	400†	850†
Modulus of elasticity (GPa)	50	45	39	200	200
Density (g/cm^3)	2.0	2.0	1.9	7.8	7.8

†Yield strength.

under field conditions. All anchor bolts were installed in a local quarry in a rock mass belonging to the Appalachian geological formation. The rock is a Greywacke with two conjugated sets of joints (dips: 70° and 20°) and random jointing near the surface. The quality of the rock mass around the installation zone is good according to both the RMR method (RMR = 75) [19] and NGI method (Q = 16–18) [20]. The *in situ* modulus of deformability, E_M, can be estimated from RMR as follows [21]: $E_M = 2\,\text{RMR} - 100\,(\text{GPa})$, so in this case, $E_M \simeq 50$ GPa. This evaluation of E_M is close to the modulus of elasticity of rock samples (61 ± 3 GPa). This local rock quarry has been involved in several field studies on grouted rock anchors at the Université de Sherbrooke [22–24].

Holes of 76.2 mm diameter were drilled vertically by a percussive and rotatory drilling process. The anchor bolts were bonded into the holes with the same type of cement grout as used in the laboratory. Two bond lengths of 150 and 450 mm were selected for the field tests which included GFRP and Dywidag bars. The top part, 150 mm above the bond length, was isolated from the grout using a PVC tube. Two anchor bolts of each bond length were tested 14 days after grouting. An overview of the anchor bolts installed in the field is shown in Fig. 4.

Figure 5 shows the set-up of the field tests. The anchor bolts were loaded using a similar method as for the ones installed in concrete blocks. The pull-out load and slip at the loaded end were monitored using a load cell and a micrometer, respectively. The load was progressively increased until reaching bolt failure.

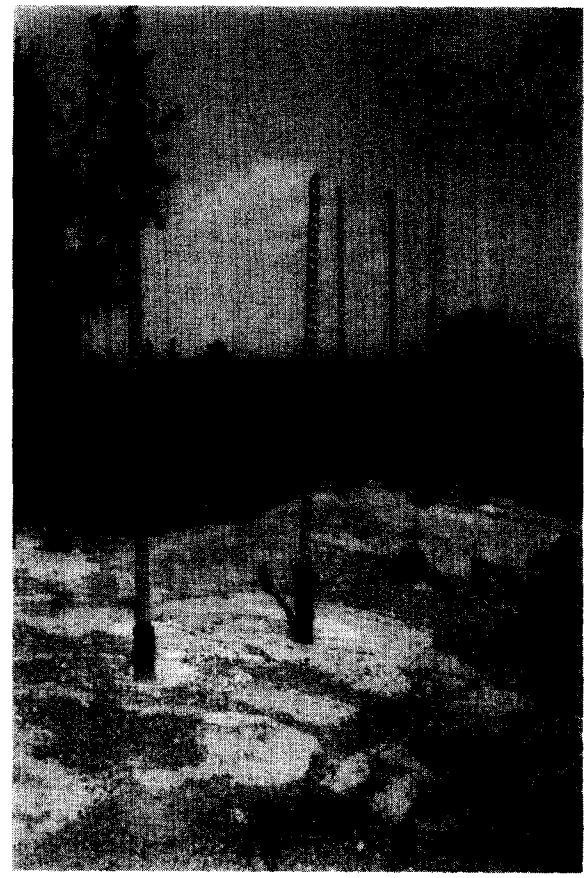

Fig. 4. Overview of anchor bolts installed in the field.

TEST RESULTS AND DISCUSSION

Load carrying capacity

The averages of maximum load carrying capacity of anchor bolts installed in concrete blocks and rock mass are shown in Figs 6 and 7, respectively.

The test results (Fig. 6) indicate that the load carrying capacity increases linearly with the increase in bond length. It also can be seen that steel bar anchor bolts obtain the highest average load carrying capacity (62 and 138 kN for 63.5 and 127 mm bond length, respectively), while GFRP C anchor bolts have the lowest average load carrying capacity (29 kN for 127 mm bond length). GFRP A and B-1 anchor bolts have an average load carrying capacity of 49 and 117 kN, and 57 and 128 kN for 63.5 and 127 mm bond length, respectively, which are

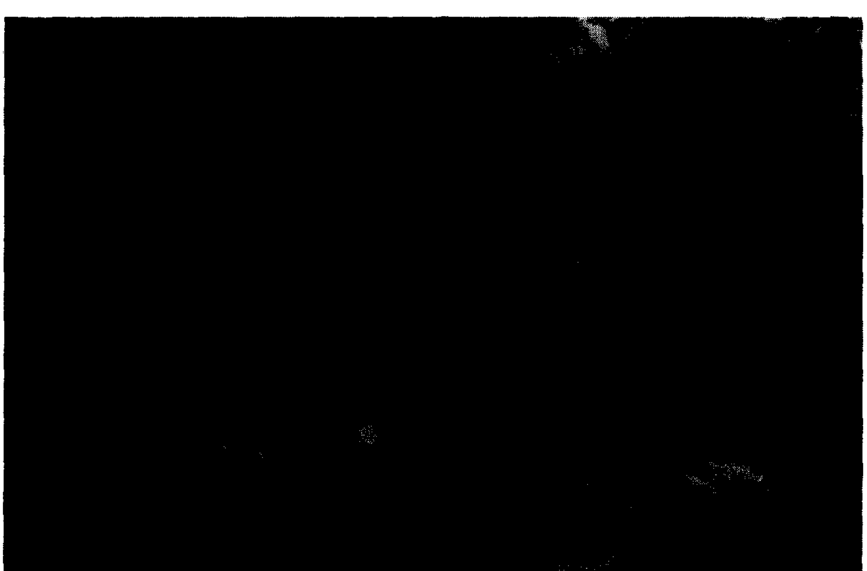

Fig. 5. Test set-up of GFRP anchor bolts in the field.

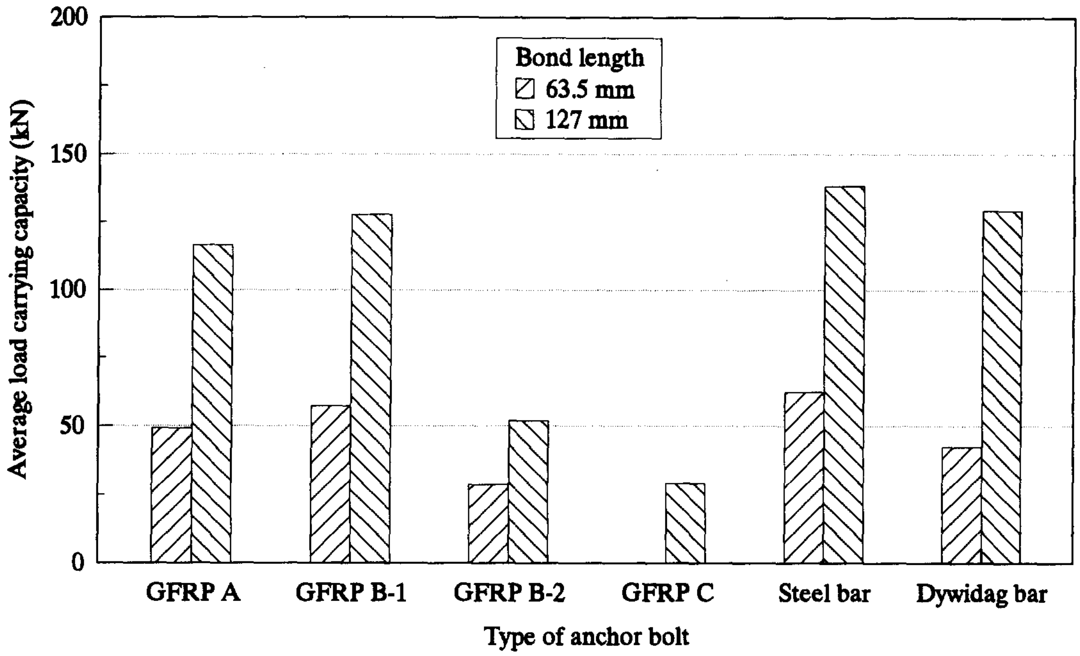

Fig. 6. Average load carrying capacity of anchor bolts installed in concrete blocks.

at the same level of Dywidag bar anchor bolts (43 and 129 kN for 63.5 and 127 mm bond length, respectively). GFRP B-2 anchor bolts have a slightly higher average load carrying capacity than GFRP C anchor bolts.

A similar pull-out behaviour to that installed in concrete blocks is observed from the field tests (Fig. 7). The load carrying capacity increases linearly as bond length increases. GFRP A and B-1 anchor bolts have an average load carrying capacity of 79 and 111 kN, respectively, for a bond length of 150 mm, which is close to that of Dywidag bar anchor bolts. GFRP C anchor bolts have the lowest load carrying capacity of 14 and 54 kN, respectively, for a bond length of 150 mm and 450 mm. In the rock mass, GFRP B-2 anchor bolts have a much higher load carrying capacity (48 and 156 kN, respectively, for a bond length of 150 and 450 mm) than that of GFRP C anchor bolts.

As expected, pull-out shear failure mechanisms occurred at the anchor–grout interface, except for GFRP A and B-1 anchor bolts with a bond length of 450 mm, which failed prematurely due to anchorage grip. The loads at bolt failure by tendon breakage were only 36% and 55% of the tensile strength of GFRP A and B-1, respectively. The failure generally occurred underneath the anchorage device, and the bars were split. Since FRP bars are sensitive to multi-axial stresses and have a low lateral shear strength and compressive strength [25], the lateral shear and compressive failure

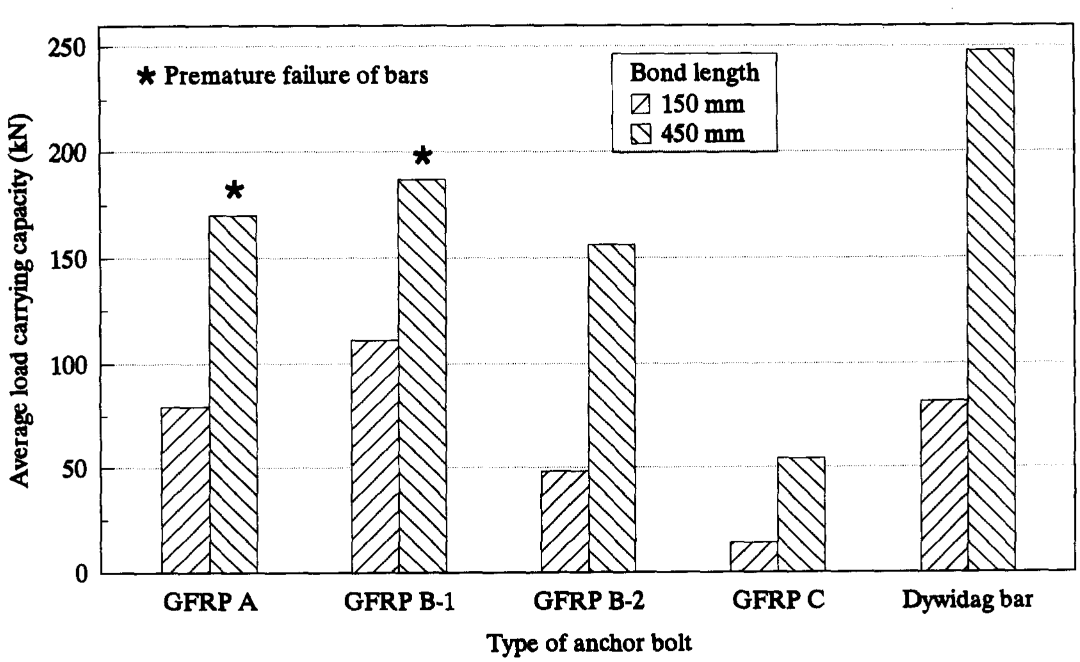

Fig. 7. Average load carrying capacity of anchor bolts installed in rock mass.

Table 4. Average bond strength and slip at failure of anchor bolts in concrete blocks

	Bond length 63.5 mm		Bond length 127 mm	
Anchor type	Bond strength (MPa)	Slip at failure (mm)	Bond strength (MPa)	Slip at failure (mm)
GFRP A	9.7 (0.1)	1.55	11.7 (1.0)	1.57
GFRP B-1	11.3 (1.1)	2.47	12.8 (2.9)	4.99
GFRP B-2	5.7 (0.1)	1.75	5.2 (0.8)	3.67
GFRP C	—	—	2.9 (0.4)	9.03
Steel bar	12.3 (0.6)	1.61	13.9 (0.5)	0.73
Dywidag	8.4 (1.6)	0.75	13.0 (1.8)	0.57

( ): standard deviation.

Table 5. Average bond strength and slip at failure of anchor bolts in rock mass

	Bond length 150 mm		Bond length 450 mm	
Anchor type	Bond strength (MPa)	Slip at failure (mm)	Bond strength (MPa)	Slip at failure (mm)
GFRP A	6.6 (0.3)	2.70	> 4.7†	—
GFRP B-1	9.3 (2.8)	2.98	> 5.2†	—
GFRP B-2	4.0 (1.2)	5.48	4.3 (0.7)	5.40
GFRP C	1.1 (0.1)	6.21	1.5 (0.6)	8.09
Dywidag	6.8 (1.4)	3.23	6.9 (1.3)	4.16

( ): standard deviation.
†Premature failure of bars.

could be the main reasons for the low failure load. This suggests that a suitable anchorage device be developed for FRP anchor bolts.

Bond strength

Tables 4 and 5 show the average shear bond strength at the grout–anchor interface at the failure of GFRP and steel anchor bolts installed in the concrete blocks and rock mass, respectively. The shear bond strength is derived by dividing the maximum pull-out load by the product of the anchor perimeter and the actual bond length.

The results shown in Tables 4 and 5 indicate that the averages of the shear bond strength of deformed GFRP A and B-1 bars are greater than those of coated (a layer of sand and resin) GFRP B-2 bars and smooth GFRP C bars. This can be explained by the fact that the deformed (or threaded) GFRP bars develop bond strength essentially by mechanical interlock mobilising the full column of cement grout in shear (bearing of the bar indentation against the grout). On the other hand, the coated and smooth bars develop their bond strength mainly by frictional shear resistance at the anchor–grout interface. This point was confirmed by visualising the failure modes at the grout–bolt interface when bars were pulled out of the grout column [13]. GFRP B-1 anchor bolts have a large deformation profile and failed due to the shearing of the grout at the grout–bolt interface. This failure mechanism is similar to steel anchor bolts. The deformation profile of GFRP A is relatively small, thus GFRP A anchor bolts failed due to partially crushing the grout and shearing at the grout–bolt interface. Consequently, the bond strength of GFRP A is lower than that of GFRP B-1 (Tables 4 and 5). GFRP B-2 failed due to the shearing of the layer of sand and resin coating on the bar. GFRP C anchor bolts failed due to the delamination of fibres at the surface and the loose fibres being sheared off. The addition of a layer of sand and resin coating in GFRP B-2 bars resulted in a decrease in bond strength (approximately 50%) compared with that of GFRP B-1 bars.

Table 6 compares the bond strength of GFRP and steel anchor bolts. It can be seen that the average bond strength of GFRP A and B-1 is generally lower (79–84% and 92%, respectively) than that of conventional steel bar bolts, but is greater (90–116% and 99–136%, respectively) than that of Dywidag bar bolts. The average bond strength of GFRP C is only 21% and 16–23% that of steel bar and Dywidag bar anchor bolts, respectively. The average bond strength of coated GFRP B-2 is between those of the deformed GFRP A and B-1 and the smooth GFRP C, 37–46% and 40–68% that of steel bar and Dywidag bar anchor bolts, respectively.

It can be noted from Tables 4 and 5 that bond strength from laboratory tests is higher than that from the field tests. This can be explained by the different radial stiffnesses at the grout–bolt interface of each host medium. Previous studies [26–28] show that this radial stiffness has a significant effect on the bond strength. The radial stiffness at the grout–bolt interface is influenced by the mechanical and elastic properties of the grout and host medium as well as the geometry of the bolt and borehole. According to the previous study [28], a calculation indicates that the host medium in the field has a lower radial stiffness than that in the laboratory due to increased grout thickness (the borehole diameter is 76.2 mm in the field compared with 50.8 mm in the laboratory). Consequently, the bond strength in the field tests is lower than that in the laboratory tests.

Table 6. Bond strength ratios of GFRP and steel anchor bolts

	Concrete block		Rock mass		
Anchor type	BL 63.5 mm	BL 127 mm	BL 150 mm	BL 450 mm	Range
GFRP A/Steel bar	0.79	0.84	—	—	0.79–0.84
GFRP B-1/Steel bar	0.92	0.92	—	—	0.92
GFRP B-2/Steel bar	0.46	0.37	—	—	0.37–0.46
GFRP C/Steel bar	—	0.21	—	—	0.21
GFRP A/Dywidag	1.16	0.90	0.97	—	0.90–1.16
GFRP B-1/Dywidag	1.35	0.99	1.36	—	0.99–1.36
GFRP B-2/Dywidag	0.68	0.40	0.59	0.62	0.40–0.68
GFRP C/Dywidag	—	0.23	0.16	0.22	0.16–0.23

BL = bond length.

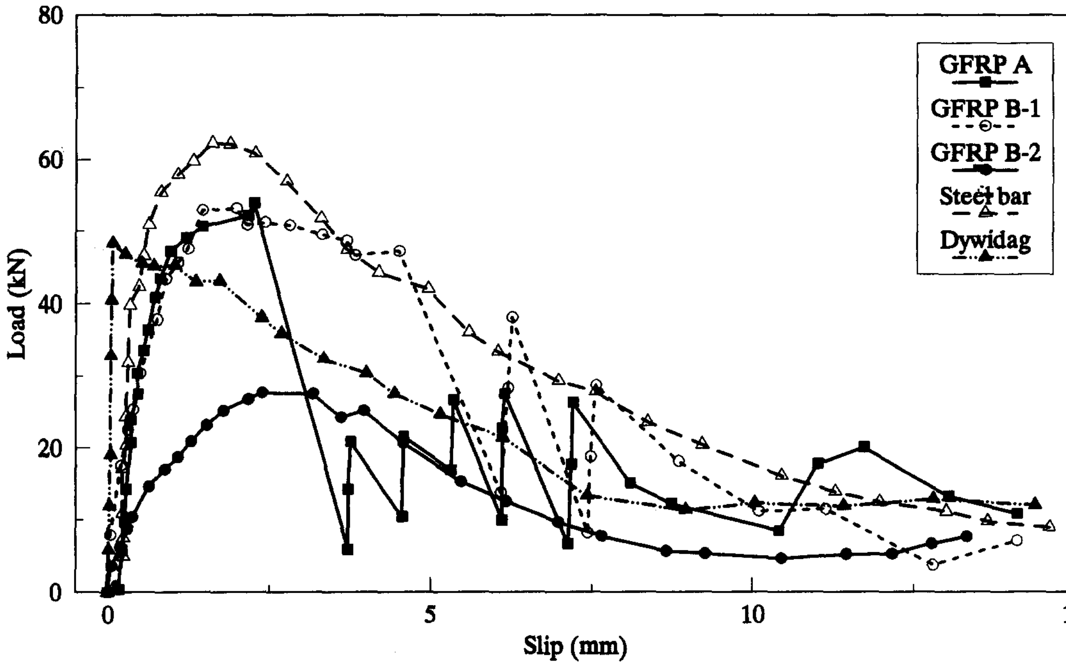

Fig. 8. Pull-out behaviour of GFRP and steel anchor bolts in concrete blocks with 63.5 mm bond length.

Load–slip behaviour

Since the modulus of elasticity of GFRP bars is approximately 25% that of steel, the elastic elongation of GFRP bars between the top of the bond length and the measuring point is four times that of steel anchor bolts. The slip along the bond length is expressed with the net displacement after taking into account the elastic elongation of GFRP bars:

$$\delta_{slip} = \delta_{total} - \delta_{bar} \quad (1)$$

where δ_{slip} = total slip of GFRP bar along the bond length; δ_{total} = measured total displacement; and δ_{bar} = GFRP bar elastic elongation.

The values of δ_{total} were measured continuously with the corresponding loads using the LVDT. The GFRP bar elongation is calculated from $\delta_{bar} = (PS)/(AE)$, where P is the applied load; A is the cross-section area; E is the modulus of elasticity; and S is the distance between the top of the bond length and the measuring point of the LVDT or micrometer.

Typical relations between the applied load and corresponding slip at the bolt head of GFRP and steel anchor bolts installed in the concrete blocks and rock mass are shown in Figs 8–11. Generally, GFRP A and B-1 anchor bolts have load–slip behaviour similar to that of steel anchor bolts. The load–slip curves comprise ascending

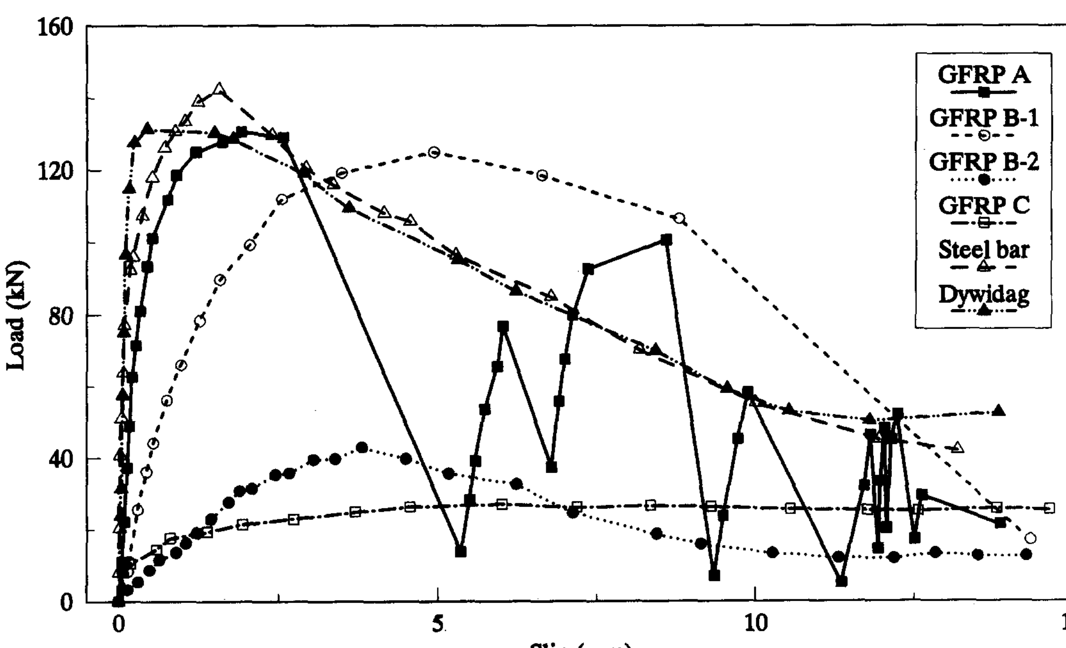

Fig. 9. Pull-out behaviour of GFRP and steel anchor bolts in concrete blocks with 127 mm bond length.

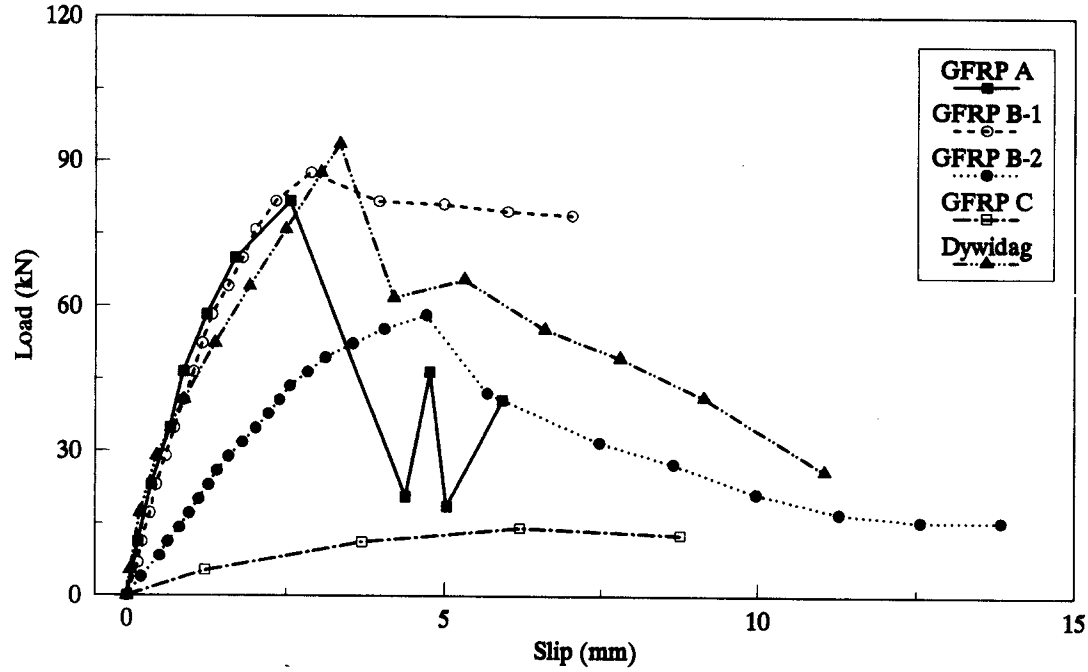

Fig. 10. Pull-out behaviour of GFRP and steel anchor bolts in rock mass with 150 mm bond length.

(pre-peak) and descending (post-peak) phases. In the ascending phase, the load increases significantly for a given increase in slip at the bolt head until the peak load is reached. The load transfer from tendon to the grout depends mainly on the adhesion and mechanical interlock. The latter contributes most to the total load for bolts with a deformed surface. GFRP B-1 anchors sustain a greater slip before reaching their peak load. In the descending phase, the load decreases sharply for a given increase in slip, and the residual load depends not only on the frictional resistance but also on the mechanical interlock between the grout and bolt as shown by the load fluctuations in Figs 8–11. The load–slip behaviour of GFRP B-2 is similar to that of GFRP A and B-1, but with a lower load amplitude and a greater slip.

GFRP C anchor bolts have a different load–slip behaviour due to the smooth surface. The load carrying capacity depends only on the adhesion between the bolt and grout prior to debonding, and on friction thereafter.

The slip at failure in Tables 4–6 is defined as the slip at the maximum load applied, and it was much greater for the GFRP anchor bolts than for steel anchor bolts. Two possible reasons are: (1) the lower modulus of elasticity, and thus the greater elastic elongation along the bond length of GFRP bolts; and (2) a greater slip of the entire bond length in GFRP bolts than in steel bolts.

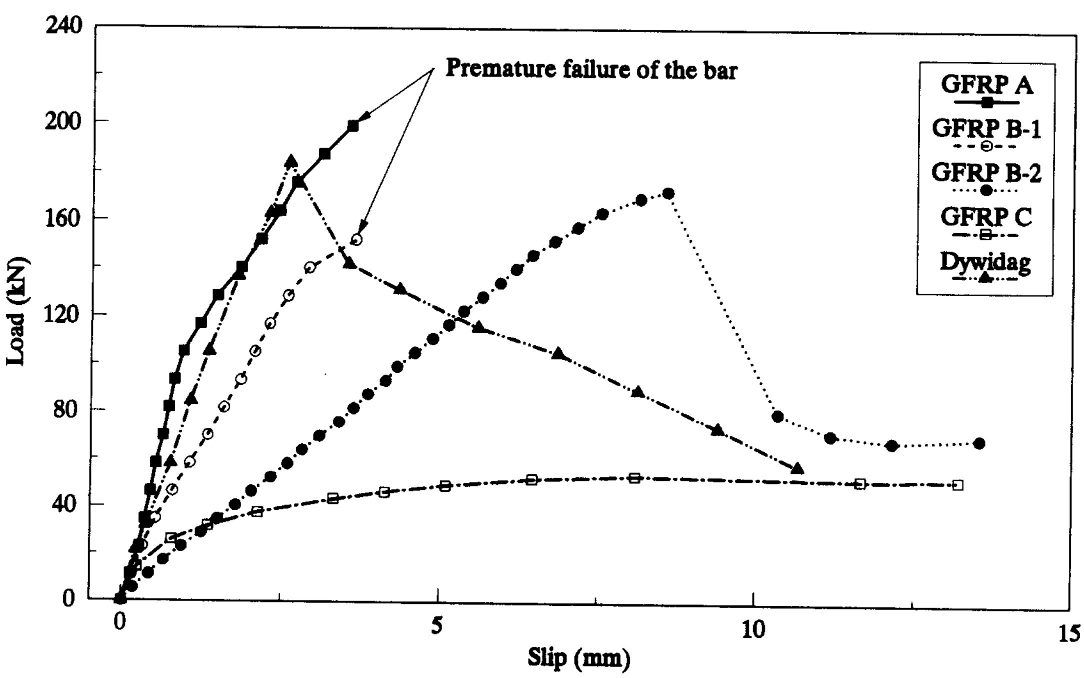

Fig. 11. Pull-out behaviour of GFRP and steel anchor bolts in rock mass with 450 mm bond length.

Table 7. Average critical bond length of GFRP and steel anchor bolts

Anchor type	Concrete block		Rock mass	
	L_c (mm)	L_c/d_b	L_c (mm)	L_c/d_b
GFRP A	570	22.4	920	36.3
GFRP B-1	365	14.3	470	18.6
GFRP B-2	805	31.7	1055	41.6
GFRP C	1305	51.3	2905	114.4
Steel bar	195	7.6	—	—
Dywidag bar	505	19.9	790	31.0

L_c: critical bond length; d_b: diameter of the anchor bolt.

Critical bond length

Results shown in Figs 6 and 7 indicate that the maximum pull-out load capacity of both GFRP and steel anchor bolts increases linearly as the bond length increases. With an excessive bond length, the bolt will fail in tension rather than being pulled out. The critical bond length is defined by the minimum embedment length necessary to bring the pull-out load to the ultimate tensile strength of the bolt, and is a function of the bond strength and ultimate tensile strength of the bar.

Based on that, the pull-out load increases linearly with bond length. The critical bond lengths of the four types of GFRP, and steel anchor bolts from the laboratory and the field tests are calculated as shown in Table 7. The critical bond length of GFRP anchor bolts varies considerably; the bond strength of GFRP anchor bolts installed in the concrete blocks is higher, whereas, the critical bond length calculated is shorter than that in the rock mass. Steel bar anchor bolts have the shortest critical bond length, while GFRP C anchor bolts have the longest critical bond length. With 1.5–2.4 times the tensile strength of steel bars, all four types of GFRP bolts have a longer bond length (1.9–6.8 times) than steel bar anchor bolts. GFRP A anchor bolts have a slightly longer critical bond length, while GFRP B-1 anchor bolts have a slightly shorter critical bond length, than Dywidag bar anchor bolts. GFRP B-2 anchor bolts have a longer critical bond length (1.3–1.6 times) than Dywidag bar anchor bolts. GFRP C anchor bolts have the longest critical bond length (6.8 times and 2.6–3.7 times that of steel and Dywidag bar anchor bolts, respectively).

CONCLUSIONS

In this investigation, experimental results on the evaluation of bond properties and pull-out behaviour of cement grouted GFRP anchor bolts were reported. Laboratory and field pull-out tests on four types of GFRP anchor bolts and two types of steel anchor bolts were conducted following the test procedures described. The GFRP anchor bolts were compared with steel anchor bolts in terms of shear strength at the anchor–grout interface, critical bond length, pull-out load–slip behaviour and failure mode. In particular, it was found that:

(1) The indentation or surface deformation produced in the GFRP bars plays a major role on the bond strength as well as on the load–slip behaviour. GFRP bars with a large deformation profile, such as GFRP B-1, give the highest bond strength and stiffness in comparison with the other GFRP bars. The addition of a layer of sand and resin coating on the surface of GFRP B-2 resulted in a decrease in both bond strength and stiffness of the anchor bolt.

(2) The bond strength of GFRP A and B-1 anchor bolts is lower than that of steel bar anchor bolts, and is slightly higher than that of Dywidag bar anchor bolts. GFRP B-2 and C anchor bolts have a significantly lower bond strength than steel anchor bolts due to the smooth surface.

(3) The critical bond length of GFRP anchor bolts varies considerably. GFRP anchor bolts have a much longer critical bond length than steel bar anchor bolts due to the higher tensile strength. GFRP A and B-1 anchor bolts have a critical bond length similar to that of Dywidag bar anchor bolts, whereas GFRP B-2 and C anchor bolts have a significantly longer critical bond length than Dywidag bar anchor bolts due to the smooth surface.

(4) GFRP A and B-1 anchor bolts have load–slip behaviour similar to that of steel anchor bolts. However, GFRP B-2 and C anchor bolts present different behaviour due to a smooth surface. The slip at failure of GFRP anchor bolts is greater than that of steel bolts mainly due to the lower modulus of elasticity.

(5) Deformed GFRP A and B-1 anchor bolts have a failure mechanism similar to steel anchor bolts, due to the crushing and shearing at the grout–bolt interface. However, the failure of GFRP C anchor bolts is due to the shear and delamination of the loose fibres on their surface, respectively.

This research has evaluated the possible application of GFRP bars as grouted anchor bolts. The test results obtained assist practising engineers in the design of GFRP anchors for a wide range of applications in civil and mining engineering, where the environments are severely aggressive, non-magnetic or non-conductive structures are required, or high corrosion will be expected.

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